

## 27. Hilbert Numbers in Ideal Theory.

J. Ber. d. DMV 34 (1925), p. 101

— E. NOETHER: Hilbert numbers in ideal theory. By Hilbert numbers one generally means numbers concerned with counting residue classes of ideals. Hilbert himself, in his “characteristic function,” gave a function that, for ideals in a polynomial domain, yields the exact number of linearly independent residue classes from a certain bound onward. For the lower degrees, Macaulay supplemented this function by a generating function that Ostrowski was able to evaluate explicitly. Macaulay also introduced counts of another kind, however, which have proved the most fundamental for general ideal theory because they admit an abstract formulation. In accordance with the general decomposition theorems, it suffices to define them for primary ideals  $q$ . The count in question is the number of residue classes that are linearly independent over the residue-class field of the associated prime ideal  $p$ . These counts split into partial systems, each determined by the number of irreducible components of  $q, q/p, q/p^2, \dots$ . The count belonging to the  $i$ -th partial system is also characterized by the number of terms in a composition series running from  $q/p^i$  to  $q/p^{i-1}$ . The full composition series and its Hilbert numbers form the counterpart, in the general case, of the representation of primary ideals as powers of prime ideals—a representation that is no longer valid, for example, in finite orders of an algebraic number field.